

# Lab07\_Teixeira

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```
library(RSQLite)
library(DBI)
```

1. Baixe o arquivo `uwmadison.sqlite3`. Conecte-se a ele usando o pacote `RSQLite`, armazenando a conexão em uma variável `conn`.

```
conn = dbConnect(SQLite(),
  '/cloud/project/database.sqlite3')
conn
```

```
<SQLiteConnection>
Path: /cloud/project/database.sqlite3
Extensions: TRUE
```

2. Quem são e quantos são os professores que lecionaram disciplinas cujo tópico era estatística (`subjects.abbreviation='STAT'`);

```
SELECT DISTINCT i.id, i.name
FROM instructors i
JOIN teachings t ON i.id = t.instructor_id
JOIN sections sec ON t.section_uuid = sec.uuid
JOIN course_offerings co ON sec.course_offering_uuid = co.uuid
JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
JOIN subjects s ON sm.subject_code = s.code
WHERE s.abbreviation = 'STAT'
```

Table 1: Displaying records 1 - 10

| id      | name        |
|---------|-------------|
| 4392906 | MINJING TAO |

| id      | name            |
|---------|-----------------|
| 393380  | DONALD PORTER   |
| 4146030 | SHENG WANG      |
| 5697552 | KUNLING HUANG   |
| 4898014 | DONG XIA        |
| 4613728 | JENNIFER NGUYEN |
| 5520811 | SEULKEE YUN     |
| 5516726 | DUZHE WANG      |
| 5696436 | YONGFENG WU     |
| 4636315 | JARED BROWN     |

```
SELECT COUNT(DISTINCT i.id) AS quantidade_professores
FROM instructors i
JOIN teachings t ON i.id = t.instructor_id
JOIN sections sec ON t.section_uuid = sec.uuid
JOIN course_offerings co ON sec.course_offering_uuid = co.uuid
JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
JOIN subjects s ON sm.subject_code = s.code
WHERE s.abbreviation = 'STAT'
```

Table 2: 1 records

| quantidade_professores |
|------------------------|
| 379                    |

3. O GPA americano é definido numa escala de 0 a 4, em que A = 4, AB = 3.5, B = 3, BC = 2.5, C = 2, D = 1 e F = 0. Determinando a nota média de cada oferecimento pela ponderação da quantidade de alunos em cada extrato com os valores numéricos de cada conceito, indique (no que se referente a disciplinas no assunto de estatística):

- Quem é o professor mais difícil?

```
SELECT
    i.id,
    i.name,
    SUM(gd.a_count*4.0 + gd.ab_count*3.5 + gd.b_count*3.0 + gd.bc_count*2.5 + gd.c_count*2.0
    SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count + gd.f
FROM instructors i
JOIN teachings t ON i.id = t.instructor_id
JOIN sections sec ON t.section_uuid = sec.uuid
```

```

JOIN course_offerings co ON sec.course_offering_uuid = co.uuid
JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
JOIN subjects s ON sm.subject_code = s.code
JOIN grade_distributions gd ON co.uuid = gd.course_offering_uuid
WHERE s.abbreviation = 'STAT'
GROUP BY i.id, i.name
HAVING SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count + gd.f_count) > 10
ORDER BY gpa_medio ASC
LIMIT 1;

```

Table 3: 1 records

| id      | name           | gpa_medio |
|---------|----------------|-----------|
| 2601633 | JAMES D KUELBS | 2.607692  |

- Quem é o professor mais fácil?

```

SELECT
    i.id,
    i.name,
    SUM(gd.a_count*4.0 + gd.ab_count*3.5 + gd.b_count*3.0 + gd.bc_count*2.5 + gd.c_count*2.0 + gd.d_count*1.5 + gd.f_count*1.0) / SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count + gd.f_count) AS gpa_medio
FROM instructors i
JOIN teachings t ON i.id = t.instructor_id
JOIN sections sec ON t.section_uuid = sec.uuid
JOIN course_offerings co ON sec.course_offering_uuid = co.uuid
JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
JOIN subjects s ON sm.subject_code = s.code
JOIN grade_distributions gd ON co.uuid = gd.course_offering_uuid
WHERE s.abbreviation = 'STAT'
GROUP BY i.id, i.name
HAVING SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count + gd.f_count) > 10
ORDER BY gpa_medio DESC
LIMIT 1;

```

Table 4: 1 records

| id      | name                         | gpa_medio |
|---------|------------------------------|-----------|
| 3884290 | SRIKANTHMADHAVAN ARAVAMUTHAN | 4         |

- Qual é a disciplina mais difícil?

```
SELECT
  c.name AS nome_disciplina,
  co.course_uuid,
  SUM(gd.a_count*4.0 + gd.ab_count*3.5 + gd.b_count*3.0 + gd.bc_count*2.5 + gd.c_count*2.0
  SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count + gd.f
FROM course_offerings co
JOIN courses c ON co.course_uuid = c.uuid
JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
JOIN subjects s ON sm.subject_code = s.code
JOIN grade_distributions gd ON co.uuid = gd.course_offering_uuid
WHERE s.abbreviation = 'STAT'
GROUP BY co.course_uuid
HAVING SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count + g
ORDER BY gpa_medio ASC
LIMIT 1;
```

Table 5: 1 records

| nome_disciplina                           | course_uuid                          | gpa_medio |
|-------------------------------------------|--------------------------------------|-----------|
| Introduction to the Theory of Probability | 3ef3ba60-426e-39ad-a61f-17d381b193d6 | 2.886374  |

- Qual é a disciplina mais fácil?

```
SELECT
  c.name AS nome_disciplina,
  co.course_uuid,
  SUM(gd.a_count*4.0 + gd.ab_count*3.5 + gd.b_count*3.0 + gd.bc_count*2.5 + gd.c_count*2.0
  SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count + gd.f
FROM course_offerings co
JOIN courses c ON co.course_uuid = c.uuid
JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
JOIN subjects s ON sm.subject_code = s.code
JOIN grade_distributions gd ON co.uuid = gd.course_offering_uuid
WHERE s.abbreviation = 'STAT'
GROUP BY co.course_uuid
HAVING SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count + g
ORDER BY gpa_medio DESC
LIMIT 1;
```

Table 6: 1 records

| nome_disciplina                | course_uuid                          | gpa_medio |
|--------------------------------|--------------------------------------|-----------|
| Empir Proc&Semiparmtrc Infernc | 99d2f325-3f56-397f-8c71-d4a2490a8e03 | 4         |

4. Desconecte do banco de dados.

```
dbDisconnect(conn)
```

```
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-09-23 16:25:36 -03"
```